

JISU JEON

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RESEARCH INTERESTS

Low temperature plasma modeling, Plasma driven solution electrochemistry,
Plasma etching and deposition for microelectronics

EDUCATION

- | | | |
|-----------------------|---|-----------------------|
| Aug. 2023 – Present | UNIVERSITY OF MICHIGAN
<i>Ph.D., Nuclear Engineering and Radiological Sciences</i> | Ann Arbor, USA |
| | <ul style="list-style-type: none">• Computational plasma science and engineering group (Principal Investigator: Mark J. Kushner)• Developed a 0D liquid chemistry framework initiated by 2D pulsed-plasma reactive gas fluxes, incorporating gas-liquid reactions at the interface and dissolutions into the bulk liquid• Formulated plasma liquid chemistries for applications of PFAS remediation and polymerization• Relevant courses: Introduction to Low Temperature Plasmas, Plasma Chemistry and Plasma Surface Interactions, Applied Machine Learning (NERS), Machine Learning (EECS, audit) | |
| Sep. 2021 – May. 2023 | SOGANG UNIVERSITY
<i>M.S., Mechanical Engineering</i> | Seoul, Korea |
| | <ul style="list-style-type: none">• Won the 4th prize at the 7th KSME-SEMES (Korean Society of Mechanical Engineers-Samsung Electronics Mechatronic Solutions) Open Innovation Challenge with a presentation on Monte Carlo collision model considering rotations of molecules to simulate sputtered film deposition• Relevant courses: Introduction to Nanotechnology and Engineering, Numerical Heat Transfer, Design of Experiments and Statistical Analyses, Classical Physics, Quantum Mechanics | |
| Mar. 2017 – Aug. 2021 | SOGANG UNIVERSITY
<i>B.S., Double major in Mechanical Engineering and Physics</i> | Seoul, Korea |
| | <ul style="list-style-type: none">• Won the 1st prize at Sogang University Student Academic Conference with a presentation on a simulation of magnetic confinement of electrons in a magnetron sputtering reactor• Relevant courses: Fluid Mechanics, Continuum Mechanics, Electromagnetism, Heat and Mass Transfer, Finite Element Analysis, Statistical Physics, Modern Optics, Physics Laboratory | |

PUBLICATION

- (In progress) [Jisu Jeon](#), Mark J. Kushner, Aniline Polymerization in Plasma Driven Solution Electrochemistry; expected to be submitted by Feb. 2027
- (In progress) [Jisu Jeon](#), Tiago C. Dias, et al., A Mechanism for Plasma Induced PFAS-in-Water Degradation on Long-Time Scales; expected to be submitted by Oct. 2026
- Tiago C. Dias, [Jisu Jeon](#), et al., Controlling Reactivity at the Plasma-Liquid Interface: Applications to Remediation of PFAS in Water, Plasma Chemistry and Plasma Processing 46, 84 (2026)

HONORS & AWARDS

- Second prize, Best PhD Oral Presentation by Organizing Committees' Choice at HAKONE XIX International Symposium on High Pressure Low Temperature Plasma Chemistry, 2026
- Fourth prize, The 7th KSME-SEMES Open Innovation Challenge, 2022
- First prize, Sogang University Student Academic Conference, 2020

SKILLS

- Python, Matlab, Fortran, C++, Pointwise, COMSOL Multiphysics®